

Safety Data Sheet

according to Regulation (EC) No. 1907/2006 (REACH)

Revision date: 31.03.2025

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SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name/designation:	Dichloromethane GPR RECTAPUR® Stabilized with about 0,002 % of 2-methyl-2-butene
Product No.:	23367
CAS No.:	75-09-2
Index No.:	602-004-00-3
EU REACH No.:	01-2119480404-41-XXXX
Other means of identification:	Methylene chloride

1.2 Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses:	General chemical reagent
Uses advised against:	The product, as such or as a component of a mixture, is not intended to be used by consumers (as defined by the REACH Regulation).

1.3 Details of the supplier of the safety data sheet

United Kingdom

VWR International Ltd.

Street	Hunter Boulevard, Magna Park
Postal code/City	Lutterworth, LE17 4XN, UK
Telephone	0800 22 33 44
Telefax	01455 55 85 86
E-mail (competent person)	SDS@avantorsciences.com

1.4 Emergency phone number

Telephone	+44 (0) 1270 502894 (CareChem24)
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SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

2.1.1 Classification according to Regulation (EC) No 1272/2008 [CLP]

Hazard classes and hazard categories	Hazard statements
Skin irritation, category 2	H315
Eye irritation, category 2	H319
Specific target organ toxicity (single exposure), category 3, narcotic effect	H336
Carcinogenicity, category 2	H351

2.2 Label elements

2.2.1 Labelling according to Regulation (EC) No. 1272/2008 [CLP]

Hazard pictograms



Signal word: Warning

Hazard statements	
H315	Causes skin irritation.
H319	Causes serious eye irritation.
H351	Suspected of causing cancer.
H336	May cause drowsiness or dizziness.

Precautionary statements	
P201	Obtain special instructions before use.
P280	Wear protective gloves/protective clothing/eye protection/face protection.
P302+P352	IF ON SKIN: Wash with plenty of water.
P304+P340	IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing
P305+P351+P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P308+P311	IF exposed or concerned: Call a POISON CENTER/doctor.

2.3 Other hazards

This substance does not meet the PBT/vPvB criteria of REACH, Annex XIII.

This product does not contain a substance that has endocrine disrupting properties.

SECTION 3: Composition / information on ingredients

3.1 Substances

Substance name:	Dichloromethane
Molecular formula:	CH ₂ Cl ₂
Molecular weight:	84.93 g/mol
CAS No.:	75-09-2
EU REACH registration No.:	01-2119480404-41-XXXX
EC No.:	200-838-9
ATE, SCL and/or M-factor:	none

SECTION 4: First aid measures

4.1 Description of first aid measures

General information

Do not leave affected person unattended. If unconscious but breathing normally, place in recovery position and seek medical advice. Take off contaminated clothing. Never give anything by mouth to an unconscious person or a person with cramps. When in doubt or if symptoms are observed, get medical advice.

After inhalation

Remove casualty to fresh air and keep warm and at rest. In case of respiratory tract irritation, consult a physician. When in doubt or if symptoms are observed, get medical advice.

In case of skin contact

Remove contaminated, saturated clothing immediately. Wash off any skin contamination immediately. In case of skin irritation, consult a physician.

After eye contact:

Rinse immediately carefully and thoroughly with eye-bath or water. Remove contact lenses, if present and easy to do. Continue rinsing. Consult an ophthalmologist.

In case of ingestion

Seek medical advice immediately (poison centre). Rinse mouth thoroughly with water.

Self-protection of the first aider

First aider: Pay attention to self-protection! Wear personal protection equipment (refer to section 8).

4.2 Most important symptoms and effects, both acute and delayed

Cough. Shortness of breath. Respiratory depression. Cardiac arrhythmias. Cardiac arrest. Pulmonary oedema. Vomiting. Dizziness. Nausea. Repeated exposure may cause skin dryness or cracking. Unconsciousness.

4.3 Indication of any immediate medical attention and special treatment needed

Substance is metabolized in the body producing carbon monoxide which increases and sustains carboxyhemoglobin levels in the blood, reducing the oxygen-carrying capacity of the blood. Administer oxygen, if necessary intubation and ventilation. In the event of severe poisoning hyperventilation should be considered. Do not administer catecholamines because of the cardiac effect caused by the product.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media

The product itself does not burn.

Co-ordinate fire-fighting measures to the fire surroundings.

Water spray.

Dry extinguishing powder.

Alcohol resistant foam.

Carbon dioxide (CO₂).

Extinguishing media which must not be used for safety reasons

Full water jet.

5.2 Special hazards arising from the substance or mixture

In case of fire and/or explosion do not breathe fumes.

Do not allow run-off from fire-fighting to enter drains or water courses.

In case of fire: Evacuate area.

In case of fire may be liberated:

Carbon monoxide

Carbon dioxide (CO₂).

Hydrogen chloride (HCl)

5.3 Advice for firefighters

Non-combustible corrosive substances (liquid).

Do not breathe gas/fumes/vapour/spray.

Fight fire with normal precautions from a reasonable distance.

Special protective equipment for firefighters:

Wear a self-contained breathing apparatus and chemical protective clothing.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

For non-emergency personnel: Wear personal protection equipment (refer to section 8). Avoid contact with eyes and skin. Do not breathe gas/fumes/vapour/spray. Remove victim out of the danger area. Stop leak if safe to do so.

6.2 Environmental precautions

Do not allow to enter into surface water or drains.

6.3 Methods and material for containment and cleaning up

Large spills: Dike or dam to contain for later disposal. Take up mechanically, placing in appropriate containers for disposal. Small spills: Absorb with liquid-binding material (sand, diatomaceous earth, acid- or universal binding agents). Dispose according to local legislation.

6.4 Reference to other sections

Personal protection equipment: see section 8 Disposal information: see section 13 Decomposition products in case of fire: see section 5.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Advices on safe handling

Wear personal protection equipment (refer to section 8).

Avoid contact with eyes and skin.

Avoid inhalation of the product.

Use extractor hood (laboratory).

Provide adequate ventilation.

Measures to prevent fire, aerosol and dust generation

Usual measures for fire prevention.

Measures required to protect the environment

Do not allow uncontrolled discharge of product into the environment.

Wash hands before breaks and after work. Avoid contact with eyes and skin. When using do not eat, drink or smoke. Provide eye shower and label its location conspicuously.

7.2 Conditions for safe storage, including any incompatibilities

Recommended storage temperature: 15°C – 25°C or 30°C depending on climatic conditions.

Storage class: 6.1D

Storage: Keep container tightly closed and in a well-ventilated place. Keep/Store only in original container. Keep cool. Protect from sunlight. Keep away from sources of heat (e.g. hot surfaces), sparks and open flames. Suitable container/equipment material: Glass Steel Stainless steel Unsuitable container/equipment material: Aluminium Polyethylene (PE) PVC (polyvinyl chloride)

7.3 Specific end use(s)

Apart from the uses mentioned in section 1.2 no other specific uses are stipulated.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Ingredient (Designation)	Source	Country	parameter	Limit value	Remark
Dichloromethane	2000/39/EC	EU	LTV	353 mg/m ³ - 100 ppm	Skin Designation
Dichloromethane	2000/39/EC	EU	STV	706 mg/m ³ - 200 ppm	Skin Designation
Dichloromethane	DNEL	EU	Worker, Dermal, long-term, systemic	12 mg/kg bw/day	
Dichloromethane	DNEL	EU	Worker, Inhalation, long-term, systemic	176 mg/m ³	
Dichloromethane	PNEC	EU	aquatic, freshwater	0.31 mg/l	Assessment factor: 20
Dichloromethane	PNEC	EU	aquatic, marine water	0.031 mg/l	Assessment factor: 200
Dichloromethane	PNEC	EU	sediment, freshwater	2.57 mg/kg	
Dichloromethane	PNEC	EU	sediment, marine water	0.26 mg/kg	
Dichloromethane	PNEC	EU	Sewage treatment plant	26 mg/l	Assessment factor: 100
Dichloromethane	PNEC	EU	soil	0.33 mg/kg	
Dichloromethane	EH40/2005 - Fourth Edition 2020	UK	LTV	353 mg/m ³ - 100 ppm	
Dichloromethane	EH40/2005 - Fourth Edition 2020	UK	STV	706 mg/m ³ - 200 ppm	

Recommended monitoring procedures:

European Standard EN 14042 (Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents)

European Standard EN 482 (Workplace exposure. Procedures for the determination of the concentration of chemical agents - Basic performance requirements)

8.2 Exposure controls

8.2.1 Appropriate engineering controls

Technical measures and the application of suitable work processes have priority over personal protection equipment. If handled uncovered, arrangements with local exhaust ventilation have to be used.

8.2.2 Personal protection equipment

Wear suitable protective clothing. When handling with chemical substances, protective clothing with CE-labels including the four control digits must be worn.

Eye/face protection

Eye glasses with side protection DIN-/EN-Norms EN 166

Recommendation: VWR 111-0432

Skin protection

When handling with chemical substances, protective gloves must be worn with the CE-label including the four control digits. Recommended glove articles DIN-/EN-Norms EN ISO 374 In the case of wanting to use the gloves again, clean them before taking off and air them well.

By short-term hand contact

Suitable material:	Butyl caoutchouc (butyl rubber)/FKM (fluoro rubber)
Thickness of the glove material:	0,70 mm
Breakthrough time:	> 120 min
Recommended glove articles:	VWR 112-3819

By long-term hand contact

Suitable material:	Butyl caoutchouc (butyl rubber)/FKM (fluoro rubber)
Thickness of the glove material:	0,70 mm
Breakthrough time:	> 120 min
Recommended glove articles:	VWR 112-3819

Respiratory protection

Respiratory protection necessary at: aerosol or mist formation

Suitable respiratory protection apparatus:	Full-/half-/quarter-face masks (EN 136/140)
Recommendation:	VWR 111-0206
Suitable material:	AXP3
Recommendation:	VWR 111-8932

Additional information

Wash hands before breaks and after work. Avoid contact with eyes and skin. When using do not eat, drink or smoke. Provide eye shower and label its location conspicuously.

8.2.3 Environmental exposure controls

no data available

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Appearance	
Physical state:	liquid
Colour:	colourless
Odour:	no data available

Safety relevant basic data

pH:	7 (20 °C)
Melting point/freezing point:	-95 °C
Initial boiling point and boiling range:	39.8 °C (1013 hPa)
Flash point:	no data available
Flammability:	Not applicable
Lower and upper explosion limit	
Lower explosion limit:	13 % (v/v)
Upper explosion limit:	22 % (v/v)
Vapour pressure:	475 hPa (20 °C)
Relative vapour density:	2.93 (20 °C)
Density and/or relative density	
Density:	1.322 g/cm ³ (20 °C)
Solubility(ies)	
Water solubility:	~20 g/l (20 °C)
Partition coefficient: n-octanol/water:	1.25 (20 °C)
Auto-ignition temperature:	605 °C (DIN 51794)
Decomposition temperature:	Not applicable
Viscosity	
Kinematic viscosity:	no data available
Dynamic viscosity:	0.43 mPa*s (20 °C)
Particle characteristics:	does not apply to liquids

9.2 Other information

Evaporation rate:	no data available
Explosive properties:	no data available
Oxidising properties:	Not applicable
Bulk density:	no data available
Refraction index:	1.4244 (589 nm; 20 °C)
Dissociation constant:	no data available
Surface tension:	no data available
Henry's Law Constant:	no data available

SECTION 10: Stability and reactivity

10.1 Reactivity

This material is non-reactive under normal conditions.

10.2 Chemical stability

The product is chemically stable under standard ambient conditions (room temperature).

10.3 Possibility of hazardous reactions

Violent reaction with:
Oxidising agent, strong.
Strong acid
Alkali (lye)
Perchlorates

10.4 Conditions to avoid

Protect from moisture.
Keep away from sources of heat (e.g. hot surfaces), sparks and open flames.
Protect from direct sunlight.
Possible decomposition might be provoked.

10.5 Incompatible materials:

Alkali metals
Aluminium
Reacts with strong oxidizing agents.

10.6 Hazardous decomposition products

no data available

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute effects

Acute oral toxicity:

Based on available data, the classification criteria are not met.

LD50: > 1600 mg/kg - Rat - (RTECS)

LDLo: > 357 mg/kg - Human - (RTECS)

Acute dermal toxicity:

Based on available data, the classification criteria are not met.

LD50: < 2000 mg/kg - Rat - (OECD 402)

Acute inhalation toxicity:

Based on available data, the classification criteria are not met.

LC50: 53 mg/l - Rat - (Japan GHS Basis for Classification Data)

Irritant and corrosive effects:

Primary irritation to the skin:

Causes skin irritation.

Irritation to eyes:

Causes serious eye irritation.

Irritation to respiratory tract:

Not applicable

Respiratory or skin sensitisation

In case of skin contact: not sensitising

After inhalation: not sensitising

STOT-single exposure

May cause drowsiness or dizziness.

STOT-repeated exposure

Not applicable

CMR effects (carcinogenicity, mutagenicity and toxicity for reproduction)**Carcinogenicity**

Suspected of causing cancer.

Germ cell mutagenicity

No indications of human germ cell mutagenicity exist.

Reproductive toxicity

No indications of human reproductive toxicity exist.

Aspiration hazard

Not applicable

Other adverse effects

no data available

Additional information

no data available

11.2 Information on other hazards

This substance does not have endocrine disrupting properties with respect to humans.

SECTION 12: Ecological information

12.1 Toxicity

Fish toxicity:

LC50: 193 - 502 mg/l (96 h) - Alexander, H.C., W.M. McCarty, and E.A. Bartlett 1978. Toxicity of Perchloroethylene, Trichloroethylene, 1,1,1-Trichloroethane, and Methylene Chloride to Fathead Minnows. Bull.Environ.Contam.Toxicol. 20(3):344-352 (OECDG Data File)

NOEC: 83 mg/l (28 d) - Pimephales promelas - ECHA

Daphnia toxicity:

EC50: 1250 - 1680 mg/l (48 h) - Bringmann, G., and F. Meinck 1964. Wassertoxikologische Beurteilung von Industrieabwassern. Gesundheits-Ingenieur 85:229-260 (OECDG Data File)

LC50: 108 - 220 mg/l (48 h) - Burton, D.T., and D.J. Fisher 1990. Acute Toxicity of...Methylene Chloride, and 2,4,6-Trichlorophenol to Juvenile Grass Shrimp and Killifish. Bull.Environ.Contam.Toxicol. 44(5):776-783

Algae toxicity:

NOEC: 550 mg/l (192 h) - Microcystis aeruginosa - ECHA

Bacteria toxicity:

EC50: 2590 mg/l (40 min) - ECHA (OECD 209)

12.2 Persistence and degradability

no data available

12.3 Bioaccumulative potential

Partition coefficient: n-octanol/water: 1.25 (20 °C)

12.4 Mobility in soil:

no data available

12.5 Results of PBT/vPvB assessment

This substance does not meet the PBT/vPvB criteria of REACH, Annex XIII.

12.6 Endocrine disrupting properties

This substance does not have endocrine disrupting properties with respect to the environment.

12.7 Other adverse effects

no data available

SECTION 13: Disposal considerations

13.1 Waste treatment methods

Appropriate disposal / Product

Dispose according to local legislation. Consult the appropriate local waste disposal expert about waste disposal.

Waste code product: 070103

Appropriate disposal / Package

Dispose according to local legislation. Handle contaminated packages in the same way as the substance itself. This material and its container must be disposed of as hazardous waste. Do not open container by force. Warning: Do not refill! Do not pierce or burn, even after use.

Additional information

European waste management legislation
Directive 2008/98/EC (Waste Framework Directive)

National waste management legislation
The Waste (England & Wales) Regulations 2011
Hazardous Waste (England & Wales) Regulations 2005

SECTION 14: Transport information

Land transport (ADR/RID)

14.1	UN number or ID number:	1593
14.2	UN proper shipping name:	DICHLOROMETHANE
14.3	Transport hazard class(es):	6.1
	Classification code:	T1
	Hazard label(s):	6.1
14.4	Packing group:	III
14.5	Environmental hazards:	No
14.6	Special precautions for user:	
	Hazard identification number (Kemler No.):	60
	Tunnel restriction code:	E
		(Passage forbidden through tunnels of category E.)

Sea transport (IMDG)

14.1	UN number or ID number:	1593
14.2	UN proper shipping name:	DICHLOROMETHANE
14.3	Transport hazard class(es):	6.1
	Classification code:	
	Hazard label(s):	6.1
14.4	Packing group:	III
14.5	Environmental hazards:	No
	Marine pollutant:	No
14.6	Special precautions for user:	
	Segregation group:	10
	EmS-No.	F-A S-A
14.7	Maritime transport in bulk according to IMO instruments	not relevant

SECTION 16: Other information

Abbreviations and acronyms

ACGIH - American Conference of Governmental Industrial Hygienists
 ADR - European Agreement concerning the International Carriage of Dangerous Goods by Road
 AGS - Committee on Hazardous Substances (Ausschuss für Gefahrstoffe)
 CLP - Regulation on Classification, Labelling and Packaging of Substances and Mixtures
 DFG - German Research Foundation (Deutsche Forschungsgemeinschaft)
 DNEL - Derived No Effect Level
 Gestis - Information system on hazardous substances of the German Social Accident Insurance (Gefahrstoffinformationssystem der Deutschen Gesetzlichen Unfallversicherung)
 IATA-DGR - International Air Transport Association-Dangerous Goods Regulations
 ICAO-TI - International Civil Aviation Organization-Technical Instructions
 IMDG - International Maritime Code for Dangerous Goods
 KOSHA - Korea Occupational Safety and Health Agency
 LTV - Long Term Value
 NIOSH - National Institute for Occupational Safety and Health
 OSHA - Occupational Safety & Health Administration
 PBT - Persistent, Bioaccumulative and Toxic
 PNEC - Predicted No Effect Concentration
 RID - Regulation concerning the International Carriage of Dangerous Goods by Rail
 STV - Short Term Value
 SVHC - Substances of Very High Concern
 vPvB - very Persistent, very Bioaccumulative

Training advice: Provide adequate information, instruction and training for operators.

Key literature references and sources for data

This Safety Data Sheet has been prepared based on information available for public as TOXNET information, European Chemicals Agency (ECHA) substance dossier, papers from international cancer research institutes (IARC Monographs), U.S. National Toxicology Program data, U.S. Agency for Toxic Substances and Disease Control (ATSDR), PubChem websites and SDS from our raw material manufacturers.

Additional information

Indication of changes Section 8: Update of exposure limit data

If you need an explanation of the change, contact the supplier (SDS@avantorsciences.com).

The above information describes exclusively the safety requirements of the product and is based on our present-day knowledge. The information is intended to give you advice about the safe handling of the product named in this safety data sheet, for storage, processing, transport and disposal. The information cannot be transferred to other products. In the case of mixing the product with other products or in the case of processing, the information on this safety data sheet is not necessarily valid for the new made-up material.